

BTEC Level 3 Applied Science · Unit 1 Biology · Worksheet**Prokaryotic and Eukaryotic Cells**

Original JDSscience practice. These questions were written for this resource and are not official exam-board items.

A Retrieval

1. What is the defining difference between prokaryotic and eukaryotic cells?
2. Name four structures of a typical bacterial cell.
3. Are viruses prokaryotic, eukaryotic, or neither? Justify.

B Knowledge

4. Compare DNA organisation in a bacterium and in an animal cell.
5. What is a plasmid, and why can it be medically important?
6. State typical size ranges for bacterial and animal cells.

C Application

7. Explain why some antibiotics can kill bacteria without killing human cells.
8. Why can bacterial populations become resistant to an antibiotic so quickly?
9. A student finds mitochondria in a cell. Which cell type is it? Explain.

D Exam-style practice

10. Give three differences between a bacterial cell and an animal cell. [3]
11. Explain how plasmids can be medically important. [3]
12. Explain why bacteria are described as prokaryotic. [2]

E Challenge

13. Mitochondria and chloroplasts contain circular DNA and 70S ribosomes. Suggest how this observation is used in the endosymbiotic theory.